What Is It a Function Of?

Name: ____________________

Position tells you where. Velocity tells you how you're moving. Everything else is what you're made of.

position	velocity	state
gravity, Coulomb	magnetic, drag	Hooke, Ohm

The question. Before any equation is written down: what is this phenomenon actually a function of?

Working rule: A law is sorted by what it depends on, not by which chapter of the textbook it lives in. Some laws resist the sort entirely, and that resistance is itself informative.

1. The transferable question

three questions, three families

Complete the mapping.

Question	Answer
Where are you?	____________
How are you moving?	____________
What are you made of?	____________

2. Sort the laws

position, velocity, or state

For each law, tick one column and name a familiar example if you can.

Law	Position	Velocity	State
Newton's gravitation			
Coulomb's law			
Magnetic (Lorentz) force			
Drag			
Coriolis force			
Hooke's law			
Ohm's law			
Ideal gas law			

Which row, if any, gave you genuine trouble? Explain why.

3. A second axis — what is being made small

Two families resist the sort above entirely: they answer a different question, what quantity is being minimised, rather than what a force depends on.

Name two investigations that belong to this second axis, and state what each one minimises.

Investigation	What is minimised

4. Where the two axes meet — interaction versus constitutive

Complete the correspondence.

Family	Kind of law
Position-dependent	____________
Velocity-dependent	____________
State-dependent	____________

5. The Coriolis problem — a law that breaks its own family

The Coriolis force is velocity-dependent by its mathematics but is not an interaction law at all. Explain why, referring to what it would take for a force to count as a genuine interaction.

Which observer feels no Coriolis force at all? What does that tell you about the difference between a force and a bookkeeping artefact of a chosen reference frame?

6. Fill the gaps — extend the taxonomy yourself

For one law not already covered above, name it, sort it, and justify the sort in one sentence.

Law	Family	Justification

7. Core results what this enquiry actually earned

The organising relations.

Family	Test question
Position-dependent	does it depend on where things are, and nothing else?
Velocity-dependent	does it vanish for anything at rest?
State-dependent	does it depend on the material, empirically, over a limited range?

Concepts earned, in order.

- A single diagnostic question, asked before any equation, does most of the sorting on its own.
- The sort cuts across every traditional chapter boundary in a physics course.
- Two investigations answer a genuinely different question, what is minimised, not what is depended on.
- The functional axis and the interaction/constitutive axis agree almost everywhere, and disagree in the one place, drag, that is most instructive to find.
- A law that is velocity-dependent by its mathematics need not be an interaction at all; the Coriolis force is the clearest case.

Every substantial claim should be accompanied by a reason, a named example, or a counterexample.